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STATE-OF-THE-ART PAPER

Design and evaluation of magnetically recoverable eggshell-based nanocomposites with high surface area for efficient Cr(VI) removal from wastewater

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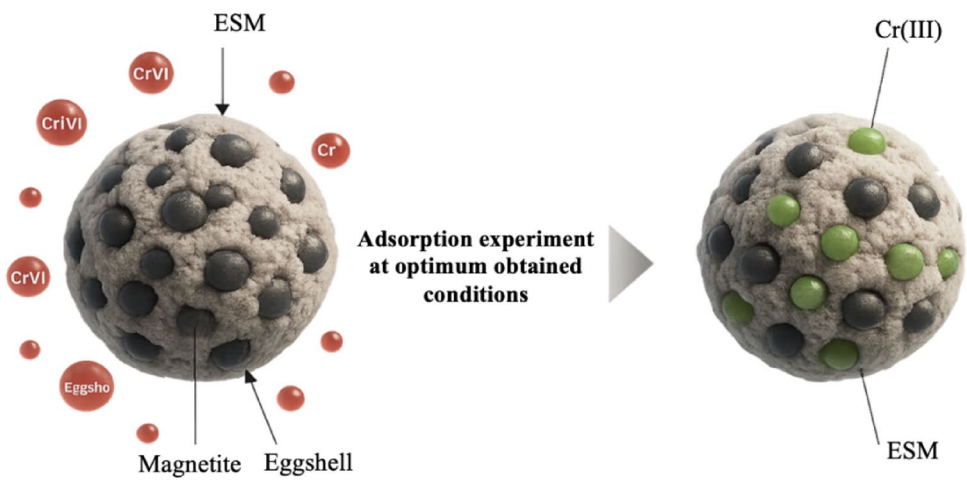
Abstract

This study introduces a sustainable eggshell–magnetite (ESM) biocomposite synthesized from waste eggshells and magnetite (Fe_3O_4) nanoparticles for efficient removal of hexavalent chromium [Cr(VI)] from aqueous solutions. The optimized ESM 2:1 (wt%) formulation achieved rapid and nearly complete Cr(VI) removal (99.05%) within 15 min at pH 3.0. Response Surface Methodology (RSM) confirmed pH and adsorbent dose as the most influential parameters, showing strong synergistic effects on adsorption capacity. Comprehensive material characterization revealed a self-activating mechanism during Cr(VI) uptake. XRD analysis demonstrated that the ESM composite retained the crystalline signatures of both calcite ($CaCO_3$) and Fe_3O_4 while exhibiting intensified calcite peaks and suppressed Fe_3O_4 reflections. BET analysis showed a 2.7-fold surface area increase after adsorption, confirming the generation of new mesoporosity. FTIR spectra evidenced shifts in Fe–O and CO_3^{2-} bands, verifying active participation of both functional groups in Cr(VI) reduction and binding. In addition, the ESM maintained over 89% removal efficiency after three regeneration cycles with an estimated treatment cost of USD 11.23 m^{-3}, demonstrating a scalable, low-cost, and circular-economy solution for decentralized wastewater treatment.

Graphical abstract



Keywords Hexavalent chromium removal · Eggshell–magnetite composite · Response surface methodology (RSM) · Adsorption kinetics · Sustainable water treatment · Waste valorization

Extended author information available on the last page of the article

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Abbreviations

ANOVA	Analysis of variance
BET	Brunauer–Emmett–Teller
BJH	Barrett–Joyner–Halenda
CCD	Central composite design
CNT	Carbon nanotubes
Cr(III)	Trivalent chromium
Cr(VI)	Hexavalent chromium
Dv(r)	BJH pore radius (volume-based)
ESM	Eggshell–Magnetite biocomposite
F. Conc.	Final concentration
FTIR	Fourier transform infrared spectroscopy
ICP-OES	Inductively coupled plasma–optical emission spectrometry
JCPDS/PDF#	Joint committee on powder diffraction standards/powder diffraction file
O&M	Operation and maintenance
PAN	Polyacrylonitrile
RSM	Response surface methodology
SEM	Scanning electron microscopy
TC	Total cost
TiO_2	Chloroprene rubber
USEPA	U.S. Environmental protection agency
VIF	Variance inflation factor
WHO	World Health Organization
XRD	X-ray diffraction

Introduction

Hexavalent chromium (Cr(VI)) remains one of the most toxic and persistent pollutants threatening aquatic ecosystems and public health [1]. Found predominantly in the form of chromate (CrO_4^{2-}), dichromate ($Cr_2O_7^{2-}$), and hydrogen chromate ($HCrO_4^-$), Cr(VI) enters water systems through industrial effluents from tanning, electroplating, textile dyeing, and metallurgy [2]. Cr(VI) is toxic and persistent pollutant that disrupt ecosystems and pose serious health risks to humans, plants, and aquatic life. Via drinking water, Cr(VI) triggers oxidative DNA damage and epigenetic alterations and injures the lungs, liver, kidneys, heart, skin, blood and immune systems, and the nervous system. These effects lead to elevated risks of cancer, multi-organ toxicity, anemia with immune dysregulation, and neurodevelopmental harm [3]. Despite international regulatory limits (0.05 mg/L set by the World Health Organization (WHO) and 0.1 mg/L by the U.S. Environmental Protection Agency (USEPA)), Cr(VI) levels in many regions, including Egypt, frequently exceed safe margins. Investigations into Nile River sediments and tributaries like the Bahr Mousa and Bahr Shebeen streams revealed high concentrations of heavy metals, with Cr(VI) contamination attributed to unregulated discharge from agriculture and industry [4, 5]. Accordingly, Cr(VI) ranked second among the detected heavy metals in fish tissues, with the order being Lead > Chromium > Cadmium > Aluminum [1, 6].

Conventional methods for Cr(VI) removal including chemical reduction, ion exchange, membrane filtration, and electrochemical precipitation have been widely studied and implemented. However, each of these approaches presents inherent limitations that hinder their broader applicability, particularly in decentralized or resource-constrained settings [6–8]. Collectively, high costs, operational complexity, and environmental trade-offs, where gains in one area can worsen another, demand decisions that balance net impacts. Among the various strategies proposed, adsorption-based methods have gained considerable attraction due to their operational simplicity, scalability, and ability to achieve high removal efficiencies even at trace-level pollutant concentrations. Within this domain, magnetite (Fe_3O_4) nanoparticles have proven as a promising class of adsorbents for Cr(VI) removal [9]. Fe_3O_4 combines strong magnetic responsiveness; facilitating easy separation and recovery with high surface area and redox activity, particularly through the Fe(II)/Fe(III) pair [10]. It can reduce Cr(VI) to the less toxic and insoluble Cr(III) [11, 12]. Nevertheless, Fe_3O_4 nanoparticles are prone to agglomeration in aqueous media, which diminishes accessible surface area and hinders performance consistency [13, 14]. Furthermore, their synthesis and stabilization often require chemical agents that are either costly or environmentally troublesome.

To overcome the limitations of single-phase adsorbents, recent research has increasingly focused on developing composite materials in which Fe_3O_4 nanoparticles are dispersed within a support matrix. Such composites improve structural stability, reusability, and surface accessibility, while preventing nanoparticle aggregation and introducing additional functional properties [15, 16]. A particularly promising and sustainable strategy involves the incorporation of biogenic waste materials as natural support matrices. Among these, eggshells (an abundant byproduct of the food industry) have gained attention due to their unique physicochemical characteristics. Primarily composed of calcium carbonate ($CaCO_3$), eggshells exhibit inherent porosity, reactive surface sites, and ion-exchange potential [17, 18]. Their alkaline nature also complements the acidic conditions typically required for Cr(VI) reduction, making them an environmentally compatible component in composite formulations. Numerous studies have demonstrated that optimizing key operational parameters, particularly pH, adsorbent dosage, temperature, and initial Cr(VI) concentration, significantly enhances removal performance [9, 19]. For example, modified biochars and fruit-peel-based biosorbents have achieved over 95% removal under acidic

conditions and extended contact times [20]. However, equilibrium times vary widely depending on adsorbent structure. While many systems require 30–180 min to reach equilibrium [21–23], Tella residue and pea (Pisum sativum) seed shell adsorbents achieved 99% Cr(VI) removal within 5 min [24]. Similarly, Sundararaman et al. [17] reported 92.9% removal using an eggshell–Fe_3O_4 nanocomposite at pH 5.5 after 66.2 min, highlighting the potential of such bioinspired materials for rapid and efficient chromium remediation.

Given Egypt's heavy dependence on textile and tanning industries, sectors characterized by intensive water use and Cr(VI)-rich effluents, this study provides a sustainable, locally accessible strategy for mitigating chromium pollution. By valorizing waste eggshells and integrating them with the magnetic and redox-active properties of Fe_3O_4, the developed composite aligns with circular economy principles while offering a scalable route toward cleaner water systems and environmental resilience. Accordingly, this research focuses on the synthesis, characterization, and performance evaluation of an ESM biocomposite engineered for the efficient removal of Cr(VI) from aqueous solutions. Anchoring Fe_3O_4 nanoparticles onto the eggshell matrix enables the composite to combine the high reactivity and reductive potential of Fe_3O_4 with the buffering, dispersive, and ion-exchange properties of $CaCO_3$, forming a synergistic system optimized for rapid and high-capacity Cr(VI) uptake. The novelty of this work lies in both the material formulation and its integrated analytical framework: a comprehensive RSM was employed to model and optimize critical process variables (pH, contact time, initial concentration, and adsorbent dose), while comprehensive physicochemical characterization (XRD, FTIR, SEM, BET) elucidated the structure–function relationship governing adsorption behavior. Furthermore, a cost analysis was performed to assess the practical feasibility and scalability of applying the ESM composite in decentralized wastewater treatment systems.

Materials and methods

Synthesis of Fe_3O_4 nanoparticles via ethylene glycol-assisted method

To synthesize Fe_3O_4 nanoparticles, 0.04 mol of ferric nitrate [$Fe(NO_3)_3 \cdot 9\ H_2O$] was dissolved in 25 mL of ethylene glycol under continuous magnetic stirring at 70 °C for 2 h. Collection of the precipitate was conducted by centrifugation, washed with ethanol and deionized water, and subjected to drying in an oven at 250 °C to remove residual solvents and moisture. The obtained xerogel was subsequently annealed at temperatures starting from 300 °C to enhance the crystalline structure and purity of the Fe_3O_4 nanoparticles.

The overall chemical reaction governing the formation of Fe_3O_4 nanoparticles can be expressed in Eq. (1).

$$3Fe(NO_3)_3 \cdot 9H_2O + 46C_2H_6O_2 \rightarrow 2Fe_3O_4 + 92CO_2 + 192H_2O + 9NO \quad (1)$$

This synthesis method ensures the production of highly stable, well-crystallized Fe_3O_4 nanoparticles with potential applications in wastewater treatment, biomedical imaging, and catalytic processes [25].

Preparation of eggshell powder for experimental applications

Waste eggshells were collected from bakeries and student restaurants from Assiut City (Egypt) as a sustainable raw material. To ensure purity, the collected eggshells were thoroughly washed multiple times with deionized water to remove impurities, organic residues, and salts that could interfere with further processing. After washing, the cleaned eggshells were dried at 100 °C for 24 h to eliminate residual moisture. The dried material was then cooled to room temperature (25 °C) before being crushed into a fine powder using a jaw crusher. The powder was sieved to obtain a uniform particle size (< 100 µm). The resulting powdered eggshell was stored in a vacuum-sealed container at room temperature to maintain its stability and prevent contamination before further use in experimental applications [26].

Preparation of eggshell-magnetite (ESM) biocomposite

Appropriate amounts of eggshell powder and magnetite nanoparticles at determined mass ratio (wt%) were weighed and thoroughly mixed using a laboratory mixer for a duration of 30 to 60 min to ensure a homogeneous distribution of magnetite within the eggshell matrix. A small amount of deionized water or ethanol was added to facilitate the mixing process. Afterward, the mixture was dried again at 80 °C to remove any residual moisture. The resulting biocomposite powder was then stored in a sealed container at room temperature until further experimental use [27].

Preparation of Cr(VI) stock solution

For the preparation of a 1000 mg/L aqueous stock solution of Cr(VI), 0.1 g of potassium dichromate ($K_2Cr_2O_7$) was precisely weighed and dissolved in 100 mL of double-distilled water. The solution was stirred until complete dissolution was achieved. Dilutions were performed to obtain

different initial Cr(VI) concentrations for adsorption studies. The prepared solutions were stored in acid-washed glass bottles and maintained at room temperature for subsequent experimental procedures.

Materials and chemicals

All materials and chemicals used in this study were of analytical grade and employed without further purification. The experimental work was entirely conducted at Assiut University, Egypt. Ferric nitrate nonahydrate [$Fe(NO_3)_3 \cdot 9\ H_2O$], used as the iron precursor for synthesizing Fe_3O_4 nanoparticles, was obtained from El-Gomhouria Chemical Supply Company (Egypt). Ethylene glycol ($C_2H_6O_2$), serving both as a solvent and reducing agent, was purchased from Sigma-Aldrich. Laboratory-grade ethanol (95%) was used in the washing steps. Potassium dichromate ($K_2Cr_2O_7$), the source of hexavalent chromium (Cr(VI)) in the adsorption studies, was acquired from ADWIC Chemicals (Egypt). Hydrochloric acid (HCl, 37%) and sodium hydroxide (NaOH) for pH adjustment were obtained from El-Nasr Chemical Company (Egypt). All aqueous solutions and rinsing steps were carried out using deionized water prepared in-house at the Faculty of Science, Assiut University.

Table 1 Experimental matrix of the central composite design (CCD) for Cr(VI) adsorption using Fe_3O_4

Run	pH	Time (min)	Initial concentration (mg/l)	Dose (mg)
1	6.5	70	10	0.5
2	6.5	120	10	0.75
3	6.5	20	10	0.75
4	10	70	10	0.75
5	6.5	70	10	1
6	3	70	10	0.75
7	6.5	70	60	1
8	6.5	20	60	0.75
9	6.5	120	60	0.75
10	3	70	60	0.75
11	10	70	60	0.75
12	6.5	70	60	0.5
13	6.5	70	35	0.75
14	3	70	35	1
15	6.5	70	35	0.75
16	6.5	120	35	0.5
17	10	70	35	1
18	3	20	35	0.75
19	10	20	35	0.75
20	6.5	20	35	1
21	6.5	20	35	0.5
22	10	120	35	0.75
23	10	70	35	0.5
24	3	120	35	0.75
25	3	70	35	0.5
26	6.5	120	35	1
27	6.5	70	35	0.75

Adsorption experiment for Cr(VI) removal

Control experiments with Fe_3O_4 using RSM

A central composite design (CCD) under the RSM framework was employed to statistically assess and optimize the key parameters affecting Cr(VI) adsorption using ESM composites. Four independent variables were considered: pH (3–10), contact time (20–120 min), initial Cr(VI) concentration (10–60 mg/L), and dosage (0.5–1.0 g/10 mL). A total of 27 experimental runs were performed, covering factorial, axial, and center points to ensure a robust quadratic model fit. All experiments were conducted in 100 mL Erlenmeyer flasks containing 10 mL of Cr(VI) solution. The pH was adjusted using 0.1 M NaOH or HCl. Fe_3O_4 dosages were accurately weighed and added to the solution, followed by orbital shaking at 200 rpm. After the predetermined contact time, the mixture was magnetically separated, and the residual Cr(VI) concentration was measured using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES). The removal efficiency (R%) was calculated using Eq. (2):

$$\%Removal = \frac{C_i - C_e}{C_i}\ x\ 100 \tag{2}$$

where C_i (mg/L) and C_e (mg/L) represent the initial and final concentrations of chromium, respectively, at a given time. All experiments were conducted in triplicate, and the average values were calculated along with the standard deviation (SD) to ensure the reliability of the results. The experimental matrix is presented in Table 1. Data were analyzed using Design-Expert via Minitab software to generate the regression model, response surface plots, and analysis of variance (ANOVA). The model's adequacy was checked via R^2, adjusted R^2, predicted R^2, and lack-of-fit tests. Optimization was performed to identify the best operating conditions for maximum Cr(VI) removal.

Adsorption study of Cr(VI) using ESM biocomposite at different ratios

The subsequent sets of experiments were conducted with the optimized conditions obtained from the previous set of Fe_3O_4 adsorbent, namely pH = 3.0, initial Cr(VI) concentration = 12.84 mg/L, and adsorbent dose = 1 g/L. In this set, 10 mL of the prepared Cr (VI) solution, followed by the addition of different ratios of ESM were investigated. Four different biosorbent compositions were assessed on weight basis (wt%): ESM (1:1) with 0.50 g of eggshell powder and 0.50 g of Fe_3O_4 nanoparticles, ESM (2:1) with 0.67 g of eggshell powder and 0.33 g of Fe_3O_4 nanoparticles, ESM

(4:1) with 0.8 g of eggshell powder and 0.2 g of Fe_3O_4 nanoparticles, ESM (5:1) with 0.84 g of eggshell powder and 0.16 g of Fe_3O_4 nanoparticles. The flasks containing the biosorbents were subjected to orbital shaking at room temperature for different contact times of 15, 40, 60, and 120 min to check the adsorption performance. After the designated time intervals, a magnetic field was applied, allowing for the separation of the magnetically ESM biosorbent from the solution. The final chromium concentration in the treated solution was determined using ICP-OES.

Recycling of ESM (2:1) biocomposite for Cr(VI) removal

As the previous experiments revealed the best performance of using ESM (2:1 wt%) for Cr(VI) removal, the reusability of the ESM biocomposite was investigated. In this set, Cr(VI) solution concentration of 12.84 mg/L was considered for three subsequent typical cycles after the main experiment. The flasks were subjected to orbital shaking at room temperature for contact times of 15, 40, 60, and 120 min. After the designated time intervals, a magnetic field was applied, allowing for the separation of the biosorbent from the solution. The biosorbent was then washed and reused for a second adsorption cycle under the same experimental conditions to assess its efficiency after recycling. The experiments were conducted in triplicate, and the average values were calculated along with the SD to ensure accuracy and reproducibility. The performance of the recycled biosorbent was analyzed, with the aim of achieving high chromium removal efficiency across all adsorption cycles.

Characterization of adsorbent

A comprehensive characterization of the ESM biocomposite was conducted using multiple analytical tech-niques to assess its morphological, structural, chemical, and textural properties. These techniques included SEM for surface morphology, FTIR for functional group identification, XRD for crystallographic analysis, and BET analysis for surface area determination.

The surface morphology and microstructural features of the samples were examined using a JEOL JSM-7500FA field emission scanning electron microscope (FE-SEM) equipped with secondary and backscattered electron detectors for high-resolution imaging and compositional contrast analysis. High-resolution images were captured at multiple magnifications to observe microstructural features before and after Cr(VI) adsorption. FTIR spectra were recorded using a Shimadzu IRAffinity-1 S FTIR Spectrophotometer in the range of 400–4000 cm^{-1}, utilizing the KBr pellet method to identify the functional groups responsible for Cr(VI) binding. XRD analysis was carried out using a PANalytical X'Pert PRO X-ray Diffractometer operating with Cu-Kα radiation (λ = 1.5406 Å) at 40 kV and 30 mA. The scanning range was set from 10° to 80° (2θ) to determine crystallinity and phase composition. BET surface area and pore structure were determined using a Quantachrome NOVA 3000e Surface Area and Pore Size Analyzer through nitrogen adsorption–desorption isotherms at liquid nitrogen temperature (−196 °C). Samples were degassed under vacuum at 120 °C for 12 h prior to analysis. The concentration of residual chromium ions was determined using an Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES, Thermo Scientific iCAP 6200 Duo), a dual-view instrument capable of simultaneous axial and radial plasma observation for precise multi-element analysis.

Results

Surface characterization and structural analysis of adsorbents

Surface textural and structural changes

As depicted in Table 2, among the individual components, Fe_3O_4 nanoparticles initially exhibited a relatively high BET surface area of 43.13 m^2/g, reflecting their nanoscale nature and well-developed morphology. After Cr(VI) adsorption, this surface area decreased to 34.02 m^2/g, indicating partial occupation of the active surface by adsorbed Cr(VI) species. The 9.11% reduction is consistent with adsorption-induced pore coverage or partial pore collapse, which has also been observed in previous magnetic nanoparticle studies used for heavy metal removal [28].

Conversely, raw eggshells initially displayed a very low BET surface area (7.04 m^2/g), characteristic of their dense, nonporous calcium carbonate matrix. However, after adsorption experiment at 15 min, the surface area increased significantly to 20.18 m^2/g. This enhancement suggests that the Cr(VI) adsorption process may have led to structural modification of the eggshell surface either through surface

Table 2 BET textural properties of Raw and Cr(VI)-loaded magnetite, eggshells, and ESM (2:1)

	Magnetite		Eggshells		ESM 2:1 (wt%)	
	Raw	After experiments	Raw	After experiments	Raw	After experiments
Surface area (m^2/g)	43.127	34.016	7.043	20.187	31.336	84.377

etching, cracking, or mineral leaching resulting in increased accessibility of internal microporous regions [29–31].

The most remarkable behavior was observed in ESM 2:1 (wt%), where the BET surface area increased from 31.34 m^2/g to 84.38 m^2/g after adsorption experiments, an increase of nearly 169%. This finding suggests that synergistic interactions between the eggshell and magnetite components may have facilitated structural rearrangement, pore opening, or exfoliation upon adsorption. Such behavior is advantageous, as it increases available active sites and suggests the composite exhibits self-activating porosity during use, enhancing its practical adsorption performance. This involves the creation of new interfacial pores or the unblocking of existing ones, rather than simple surface etching [32].

Nitrogen adsorption–desorption isotherms were recorded at −196 °C for magnetite nanoparticles, raw eggshell powder, and the ESM (2:1 wt%) biocomposite before and after Cr(VI) adsorption. The obtained curves are presented in Figure S1 which were interpreted following the International Union of Pure and Applied Chemistry (IUPAC) classification of physisorption isotherms and hysteresis loops.

The raw Fe_3O_4 nanoparticles exhibited a typical Type IV isotherm with an H2(b)-type hysteresis loop, indicating a predominately mesoporous structure associated with ink-bottle-shaped pores and non-uniform pore connectivity. The distinct hysteresis between relative pressures $(P/P_o) = 0.4$–0.9 signifies capillary condensation within mesopores. After Cr(VI) adsorption, the overall N_2 uptake decreased and the hysteresis loop became narrower, reflecting partial blockage of the pore channels by adsorbed chromium species and a decrease in available surface area (from 43.13 to 34.02 $m^2\ g^{-1}$). These changes confirm that the Cr(VI) ions are physically and chemically retained within the mesoporous framework, in agreement with the surface saturation effects observed in BET results (Table 2).

In contrast, the raw eggshell powder showed a Type II isotherm characteristic of nonporous or macroporous materials with unrestricted multilayer adsorption. The absence of a pronounced hysteresis loop confirms the dominance of large pores and external surface adsorption. Interestingly, after Cr(VI) adsorption, the curve gradually transformed toward a Type IV isotherm with a weak H3-type hysteresis loop. This behavior suggests that the adsorption process induced structural etching and development of slit-like pores on the calcite surface, as also supported by the threefold increase in surface area (from 7.04 to 20.18 $m^2\ g^{-1}$). Such surface activation is attributed to partial leaching and re-crystallization phenomena during acidic exposure and Cr(VI) binding [30].

The ESM (2:1 wt%) biocomposite displayed a hybrid Type IV isotherm with a well-defined H3-type hysteresis loop, confirming the coexistence of mesoporous and slit-shaped pore structures derived from the composite architecture. Before adsorption, the composite exhibited a steep rise in adsorption volume at relative pressures between 0.6 and 0.9, signifying capillary condensation in mesopores and uniform pore distribution. Following Cr(VI) adsorption, the total N_2 uptake markedly increased and the hysteresis loop widened, indicating further development of open mesoporous channels and enhanced interparticle voids. The BJH analysis corroborated this observation by showing an increase in average pore volume from 0.039 to 0.100 $cm^3\ g^{-1}$ and a nearly stable mean pore radius (~ 19.9 Å), implying that the composite underwent adsorption-induced pore activation rather than collapse [33]. This self-activating behavior can be ascribed to synergistic structural rearrangement between the eggshell and Fe_3O_4 phases, producing new accessible adsorption sites and validating the significant high Cr(VI) removal efficiency (99.05%), as the hydrated ions can more easily enter and interact with internal active sites [34, 35].

Accordingly, the IUPAC-based classification reveals that all materials are predominantly mesoporous, but their textural evolutions differ upon interaction with Cr(VI). Fe_3O_4 experiences partial pore blockage (Type IV to narrower H2 loop), eggshell undergoes pore formation (Type II to weak Type IV/H3), whereas the ESM composite exhibits enhanced mesoporosity and hysteresis widening (Type IV/H3). These observations confirm that the synergistic combination of magnetite and eggshell not only preserves but amplifies the mesoporous framework, enabling rapid diffusion and strong retention of Cr(VI) ions.

SEM analysis

SEM was conducted to investigate the surface morphology and microstructural changes of the raw and Cr(VI)-loaded different adsorbents. Images shown in Fig. S2 were obtained at magnifications of ×5,000, ×10,000, and ×20,000, allowing detailed visualization of particle texture, porosity, and surface transformations that occurred due to adsorption. According to Fig. S2, SEM images of raw Fe_3O_4 show spherical to semi-spherical particles with moderately rough surfaces and partial aggregation. At higher magnifications (×20,000), the particles exhibit a nanoscale texture with visible inter-particle voids, which support their initial high BET surface area. Following peak Cr(VI) adsorption, the Fe_3O_4 surface appears smoother and more compact, suggesting partial pore blockage and surface saturation due to adsorbed chromium species. This morphological change aligns with the observed drop in BET surface area to 34.02 m^2/g and supports the reduction in available active surface area post-treatment [36].

Raw eggshells reveal a dense, layered surface structure with minimal porosity, typical of calcium carbonate-rich biomaterials. The surface appears relatively featureless at ×5000, with slight textural variation emerging at higher magnification. After peak Cr(VI) adsorption, a significant transformation is clear: the surface becomes more irregular and porous, with signs of etching, cracking, and particle erosion. These structural changes corroborate the notable increase in BET surface area (from 7.04 to 20.18 m^2/g) and pore volume, indicating that the adsorption process may have induced partial mineral dissolution or disruption of the calcitic matrix, thus enhancing internal porosity [14, 37].

The raw ESM 2:1 composite presents a heterogeneous surface consisting of fine Fe_3O_4 nanoparticles dispersed over the larger, dense eggshell matrix. This configuration offers both micro- and meso-scale surface features and confirms effective mixing of the two phases. After Cr(VI) uptake, SEM images reveal a more fragmented and rugged surface, with noticeable textural deepening and exposure of inner layers, possibly due to structural rearrangement or localized weakening induced by chemical interactions. The visual roughness and expanded microstructure are in strong agreement with the dramatic increase in BET surface area (from 31.34 to 84.38 m^2/g) and support the idea that adsorption triggered a self-activation or surface-opening phenomenon [38]. Moreover, the post-adsorption images of the composite suggest a retained overall structural framework. Unlike many biosorbents that collapse or flake upon reuse, the composite appears morphologically stable, with maintained surface complexity, further underscoring its suitability for repeated Cr(VI) removal operations.

FTIR analysis

FTIR spectroscopy provides critical information about the functional groups and chemical interactions involved in Cr(VI) adsorption by the different adsorbents. By comparing spectra before and after adsorption (Fig. S3), it is possible to elucidate the mechanisms and active sites responsible for Cr(VI) binding.

Regarding the Fe_3O_4 nanoparticles before adsorption, characteristic Fe–O stretching vibrations were observed at 560 cm^{-1} and 501 cm^{-1}, confirming the presence of Fe_3O_4. Broad bands around 3423 cm^{-1} indicate surface –OH groups, which can participate in hydrogen bonding and surface reactions [39]. Post-adsorption spectra show broadening and a slight shift of the O–H band and the Fe–O peaks (from 560 to ~ 598 cm^{-1}), suggesting enhanced hydrogen bonding and direct Fe–O involvement in Cr(VI) interaction. These shifts indicate that both surface hydroxyl and Fe–O sites play a role in adsorption and possible Cr(VI) reduction to Cr(III), consistent with the Fe(II)/Fe(III) redox activity of Fe_3O_4 [17]. In addition, the persistence of Fe–O bands indicates that the Fe_3O_4 structure remains stable after adsorption.

Raw eggshells display strong carbonate (CO_3^{2-}) bands near 1420–1450 cm^{-1}, 875 cm^{-1}, and 712 cm^{-1}, confirming the calcite phase (Fig. S3). Weak C = O/CO_2 overtone near 1798 cm^{-1} was also observed. O–H and N–H stretching bands appear around 3400 cm^{-1}, reflecting the presence of surface hydroxyl and organic residues from the eggshell membrane. After adsorption, shifts in carbonate peaks (1420 to 1427 cm^{-1}) and reduced O–H band intensity suggest that these groups are directly involved in Cr(VI) binding, likely through ion exchange or surface complexation [40]. The appearance of a weak band near 630 cm^{-1} indicates minor Fe–O vibration due to trace iron impregnation or surface interactions after exposure [41].

The ESM (2:1 wt%) biocomposite spectra (Fig. S3) combines features of both eggshell (carbonate and O–H bands) and Fe_3O_4 (Fe–O bands), confirming successful integration [17]. Before adsorption, the main absorption peaks appear at 3423 cm^{-1} (O–H), 2924 cm^{-1} (C–H), 1356 cm^{-1} (CO_3^{2-} bending), 1042 cm^{-1} (C–O or Si–O stretching), and 598–483 cm^{-1} (Fe–O lattice vibrations). After Cr(VI) adsorption, significant broadening of the O–H band, slight shifts in carbonate peaks, and intensified Fe–O vibrations are observed. These variations indicate that both carbonate and iron oxide groups participate actively in Cr(VI) binding. Such spectral modifications support a synergistic adsorption mechanism where the eggshell matrix enhances Fe_3O_4 dispersion and accessibility, while Fe(II) sites facilitate Cr(VI) reduction to Cr(III) [42, 43].

The combined spectral evidence confirms that –OH, CO_3^{2-}, and Fe–O functional groups are central to Cr(VI) adsorption in the investigated materials. Therefore, FTIR results align well with BET and XRD findings, confirming surface activation and chemical stability of the ESM composite after Cr(VI) adsorption.

XRD analysis

To elucidate the structural evolution of the individual components and their composite before and after Cr(VI) adsorption, XRD analyses were performed (Fig. S4). The diffraction patterns clearly reveal the crystalline phases present in Fe_3O_4 nanoparticles, eggshells, and the ESM composite and provide direct evidence of the phase stability and transformation mechanisms governing Cr(VI) removal (Table 3).

The Fe_3O_4 sample (Fig. S4-a) shows several broad but distinct reflections at $2\theta \approx 30.1°$, 35.5°, 43.1°, 53.4°, 57.0°, and 62.6°, corresponding respectively to the (220), (311), (400), (422), (511), and (440) planes of cubic magnetite

Table 3 Summary of XRD phase identification, corresponding JCPDS reference numbers, and structural behavior of Fe_3O_4, eggshells, and ESM composite before and after Cr(VI) adsorption

Adsorbent	Dominant phase	JCPDS No.	Key 2θ (°) / (hkl)	Behavior after Cr(VI) adsorption
Fe_3O_4	Magnetite (Fe_3O_4)	19–0629	30.1° (220), 35.5° (311), 43.1° (400), 57.0° (511), 62.6° (440)	Peaks persist with additional sharp lines from Fe–Cr precipitate; confirms reductive-precipitation mechanism.
Eggshell	Calcite ($CaCO_3$)	05-0586	23.0° (012), 29.4° (104), 31.4° (006), 36.0° (110), 39.4° (113), 43.1° (202)	Peaks sharpen and intensify; increased crystallinity and porosity via surface activation.
ESM (2:1 wt%) composite	Calcite + Magnetite	05-0586 + 19–0629	Combined reflections at 29.4° ($CaCO_3$, 104) and 35.5° (Fe_3O_4, 311)	Calcite reflections dominate; magnetite peaks subdued; no Fe–Cr phase detected; framework preserved.

(JCPDS 19–0629) [44]. The relatively low peak sharpness confirms the nanocrystalline nature of the synthesized Fe_3O_4, which is consistent with its high initial BET surface area (43.13 m^2 g^{-1}) and the uniform nanoscale morphology observed in SEM [45]. After Cr(VI) adsorption (Figure S4-b), the diffraction pattern transforms markedly. All characteristic reflections of Fe_3O_4 remain, confirming structural stability; however, new sharp peaks appear, indicating the formation of an additional, more crystalline phase. This newly formed crystalline structure can be attributed to a solid-state reductive-precipitation process in which Cr(VI) is reduced to Cr(III) by surface Fe(II) sites and co-precipitates with Fe(III) to yield a mixed (Fe, Cr)-oxide or spinel-like phase [44, 46, 47]. This transformation explains the smoother particle morphology and reduced surface area (from 43.13 to 34.02 m^2 g^{-1}) observed by SEM and BET, which result from pore filling and surface densification caused by the newly formed precipitate layer.

The XRD pattern of the raw eggshell powder (Fig. S4-c) is dominated by intense, sharp reflections at $2\theta \approx 23.0°$, 29.4°, 31.4°, 36.0°, 39.4°, 43.1°, 47.5°, 48.5°, 57.3°, and 60.5°, which correspond to the (012), (104), (006), (110), (113), (202), (018), (116), (122), and (214) planes of trigonal calcite $CaCO_3$ (JCPDS 05-0586). This confirms the eggshells consist almost entirely of crystalline calcite, explaining their dense surface and low BET surface area (7.04 m^2 g^{-1}). Following Cr(VI) adsorption (Fig. S4-d), the calcite reflections become sharper and more intense, especially the major (104) peak at 29.4°, indicating increased crystallinity and lattice ordering. No new crystalline phases are observed, suggesting that the adsorption process does not alter the fundamental $CaCO_3$ framework. Instead, the enhanced peak sharpness and increased BET area (20.18 m^2 g^{-1}) indicate a dissolution–reprecipitation or surface-activation process [38]. The partial dissolution of $CaCO_3$ under slightly acidic adsorption conditions likely exposes fresh crystalline surfaces and generates new mesopores, consistent with the type IV isotherm and H_3 hysteresis loop observed in BET [48]. This structural rearrangement explains the improvement in both crystallinity and porosity, while FTIR results confirm that carbonate groups (CO_3^{2-}) actively participate in surface complexation with Cr(VI)/Cr(III) species.

The XRD pattern of the ESM composite before adsorption (Fig. S4-e) exhibits a clear superposition of both Fe_3O_4 (JCPDS 19–0629) and $CaCO_3$ (JCPDS 05-0586) reflections. Sharp peaks at 29.4° and 36.0° arise from calcite's (104) and (110) planes, while the broader, lower-intensity reflections at 30.1°, 35.5°, 43.1°, and 57.0° correspond to Fe_3O_4's (220), (311), (400), and (511) planes. The coexistence of these two crystalline phases confirms the successful incorporation of Fe_3O_4 nanoparticles within the eggshell matrix. The intermediate surface area (31.34 m^2 g^{-1}) supports this composite structure, reflecting the heterogeneous texture observed by SEM. After Cr(VI) adsorption (Fig. S4-f), the diffraction pattern becomes dominated by calcite reflections (JCPDS 05-0586), with notably sharpened and intensified peaks, whereas the Fe_3O_4 peaks are weak or partially masked. No additional reflections corresponding to Fe–Cr oxides or chromite spinels (e.g., $FeCr_2O_4$ 34–0140 or Cr_2O_3 38-1479) are detected, indicating that any Fe–Cr products formed are amorphous or nanoscopic [49]. The eggshell matrix evidently provides a stabilizing, buffering environment that prevents large-scale crystallization of Fe–Cr species and preserves the overall framework. The enhanced crystallinity of the calcite phase, along with the absence of new crystalline by-products, confirms that the composite structure remains intact and that the reaction proceeds primarily through synergistic interactions between the two phases. This behavior complements the BET and SEM results, which show a tripling of surface area (to 84.38 m^2 g^{-1}) and the emergence of a more porous and activated texture. Accordingly, the Cr(VI) removal efficiency is therefore attributed to a cooperative mechanism: Fe_3O_4 nanoparticles provide the reductive sites for Cr(VI) to Cr(III) conversion, while the $CaCO_3$ matrix undergoes structural reordering that amplifies mesoporosity and diffusion capacity, ensuring efficient ion capture and retention.

Table 4 Coded regression coefficients, standard errors, t-values, *p*-values, and variance inflation factors (VIF) for the response surface model of Cr(VI) removal

Term	Coef	SE Coef	T-Value	*P*-Value	VIF
Constant	1459	330	4.43	0.001	
pH	24.4	14.1	1.73	0.110	2188.00
Time	35.8	14.1	2.54	0.026	2188.00
Conc.	67.8	14.1	4.80	0.000	2188.00
Dose	1072	244	4.39	0.001	6562.00
pH*pH	2.682	0.452	5.93	0.000	1.25
Time*Time	−0.136	0.452	−0.30	0.770	1.25
Conc.*Conc.	2.381	0.452	5.26	0.000	1.25
Dose*Dose	197.3	45.2	4.36	0.001	6562.25
pH*Time	0.401	0.522	0.77	0.458	1.00
pH*Conc.	−0.905	0.522	−1.73	0.109	1.00
pH*Dose	8.66	5.22	1.66	0.123	2188.00
Time*Conc.	−1.238	0.522	−2.37	0.035	1.00
Time*Dose	13.79	5.22	2.64	0.022	2188.00
Conc.*Dose	23.72	5.22	4.54	0.001	2188.00

Table 5 ANOVA for the response surface quadratic model evaluating the effects of pH, contact time, initial Cr(VI) concentration, and Fe_3O_4 dose on final Cr(VI) concentration

Source	DF	Adj SS	Adj MS	F-Value	*P*-Value
Model	14	321.623	22.9731	21.04	0.000
Linear	4	56.527	14.1316	12.95	0.000
pH	1	3.259	3.2595	2.99	0.110
Time	1	7.037	7.0372	6.45	0.026
Conc.	1	25.196	25.1957	23.08	0.000
Dose	1	21.034	21.0341	19.27	0.001
Square	4	70.699	17.6749	16.19	0.000
pH*pH	1	38.363	38.3630	35.14	0.000
Time*Time	1	0.098	0.0979	0.09	0.770
Conc.*Conc.	1	30.229	30.2292	27.69	0.000
Dose*Dose	1	20.772	20.7717	19.03	0.001
2-Way Interaction	6	43.158	7.1930	6.59	0.003
pH*Time	1	0.642	0.6424	0.59	0.458
pH*Conc.	1	3.272	3.2725	3.00	0.109
pH*Dose	1	2.996	2.9964	2.74	0.123
Time*Conc.	1	6.128	6.1281	5.61	0.035
Time*Dose	1	7.604	7.6038	6.97	0.022
Conc.*Dose	1	22.515	22.5146	20.62	0.001
Error	12	13.100	1.0917		
Lack-of-Fit	10	12.954	1.2954	17.73	0.055
Pure error	2	0.146	0.0730		
Total	26	334.723			

RSM experimental results for Cr(VI) removal using Fe_3O_4 only

The regression analysis of the experimental data revealed a statistically robust second-order polynomial model describing the relationship between final Cr(VI) concentration and four independent process variables: pH, contact time, initial Cr(VI) concentration, and Fe_3O_4 nano adsorbent dose (Tables 4 and 5). The model's high coefficient of determination ($R^2 = 96.09\%$) indicates that over 96% of the variability in the final Cr(VI) concentration is explained by the regression equation. The adjusted R^2 value of 91.52% confirms that the model remains highly explanatory even when accounting for the number of predictors, while the predicted R^2 of 77.61% suggests good predictive capability. The residual standard deviation (S = 1.0448) reflects a narrow spread of residuals, and the model's ANOVA results reinforce its statistical significance with an overall p-value of 0.000 and an F-value of 21.04, confirming that the model as a whole reliably captures the behavior of the system.

Furthermore, the linear terms (−3.098 × pH, −0.0835 × Time, −0.2659 × Conc., and −72.1 × Dose) present the direct effects of each variable. All coefficients are negative, indicating that increasing any of these parameters leads to a reduction in final Cr(VI) concentration and thereby enhances removal efficiency. Among them, Fe_3O_4 dose is the most influential variable in uncoded form, with a steep coefficient of −72.1. This reflects the critical role of adsorbent dosage in providing sufficient active sites for Cr(VI) binding. pH also exhibits a significant negative linear effect, implying that acidic conditions favor Cr(VI) removal, most likely due to enhanced electrostatic attraction between negatively charged chromate ions and the protonated Fe_3O_4 surface. Although the effects of time and concentration are less intense in the uncoded regression, they remain directionally favorable toward improving removal.

The quadratic terms (pH^2, $Time^2$, $Conc.^2$, and $Dose^2$) capture curvature in the response surface, suggesting that the system's behavior is not purely linear. The positive coefficients for pH^2 (+0.2189), $Conc^2$ (+0.003809), and $Dose^2$ (+31.58) indicate the existence of minimum points within the experimental ranges of these variables. This means that after reaching a certain optimal value, further increases in these parameters may actually lead to reduced removal efficiency. For example, overdosing beyond the optimal range can cause agglomeration of Fe_3O_4 particles, decreasing their effective surface area. The large coefficient of $dose^2$ also underscores the sensitivity of Cr(VI) removal to dosage optimization. Conversely, $Time^2$ had a small and negative coefficient (−0.000054), indicating a nearly linear effect over the tested duration; increasing contact time continues to enhance adsorption but with diminishing incremental benefits.

$$\begin{aligned} \text{F. Conc.} = {} & 42.75 - 3.098\ \text{pH} - 0.0835\ \text{Time} \\ & - 0.2659\ \text{Conc.} \\ & - 72.1\ \text{Dose} + 0.2189\ \text{pH} * \text{pH} \\ & - 0.000054\ \text{Time} * \text{Time} \\ & + 0.003809\ \text{Conc.} * \text{Conc.} + 31.58\ \text{Dose} * \text{Dose} \\ & + 0.00229\ \text{pH} * \text{Time} \\ & - 0.01034\ \text{pH} * \text{Conc.} + 0.989\ \text{pH} * \text{Dose} \\ & - 0.000990\ \text{Time} * \text{Conc.} + 0.1103\ \text{Time} * \text{Dose} \\ & + 0.3796\ \text{Conc.} * \text{Dose} \end{aligned} \tag{3}$$

F. Conc.: the final concentration of Cr((VI), Conc.: the initial concentration

Figure 1 illustrates how pairs of variables influence system performance. A joint test showed that the interaction effects were significant ($p = 0.003$), although several pairwise interactions did not reach individual statistical significance. Much clearer separation of response curves is seen in the Time × Dose and Concentration × Dose plots, reflecting strong synergistic behavior. The most notable was the Concentration × Dose interaction, with a p-value of 0.001, indicating a strong compensatory relationship where increasing the dose offsets the negative effects of high initial Cr(VI) concentrations. Similarly, Time × Dose ($p = 0.022$) and Time × Concentration ($p = 0.035$) interactions were significant, suggesting that the benefit of longer contact time is magnified under specific concentrations or dosing conditions. At higher doses, extended contact time significantly enhances removal, while higher doses also counteract the negative impact of high initial concentrations, confirming the essential role of dose optimization in real applications.

In contrast, in the pH × Time and pH × Concentration plots, the near-parallel lines reflect the absence of meaningful interaction between these variables. Similarly, the pH × Dose interaction plot reveals slight divergence, implying a modest combined effect. These interactions were not statistically significant, indicating that pH's effects, while important, do not strongly modify or depend on other variables in this system. While pH does influence Cr(VI) removal, its effect is independent of changes in time or concentration.

As shown in Figs. 2 and 3, contour and surface plots add visual clarity to these interactions and nonlinearities. Each interaction subplot presents the combined effect of two variables on final Cr(VI) concentration while keeping the other two variables constant. These plots effectively highlight optimal operating regions for Cr(VI) removal and illustrate zones where synergistic behavior between factors enhances treatment performance.

The corresponding 3D surface plots shown in Fig. 3 provide a visual representation of the same two-factor interactions, revealing the curvature of the response surface and further clarifying the optimal conditions for maximum Cr(VI) removal efficiency. The pH vs. Time and pH vs. Concentration plots both confirm that acidic conditions (pH 3–4) are optimal for minimizing final Cr(VI) levels. Similarly, surface plots of Time vs. Dose and Concentration vs. Dose show valleys and slopes that reflect high removal efficiency

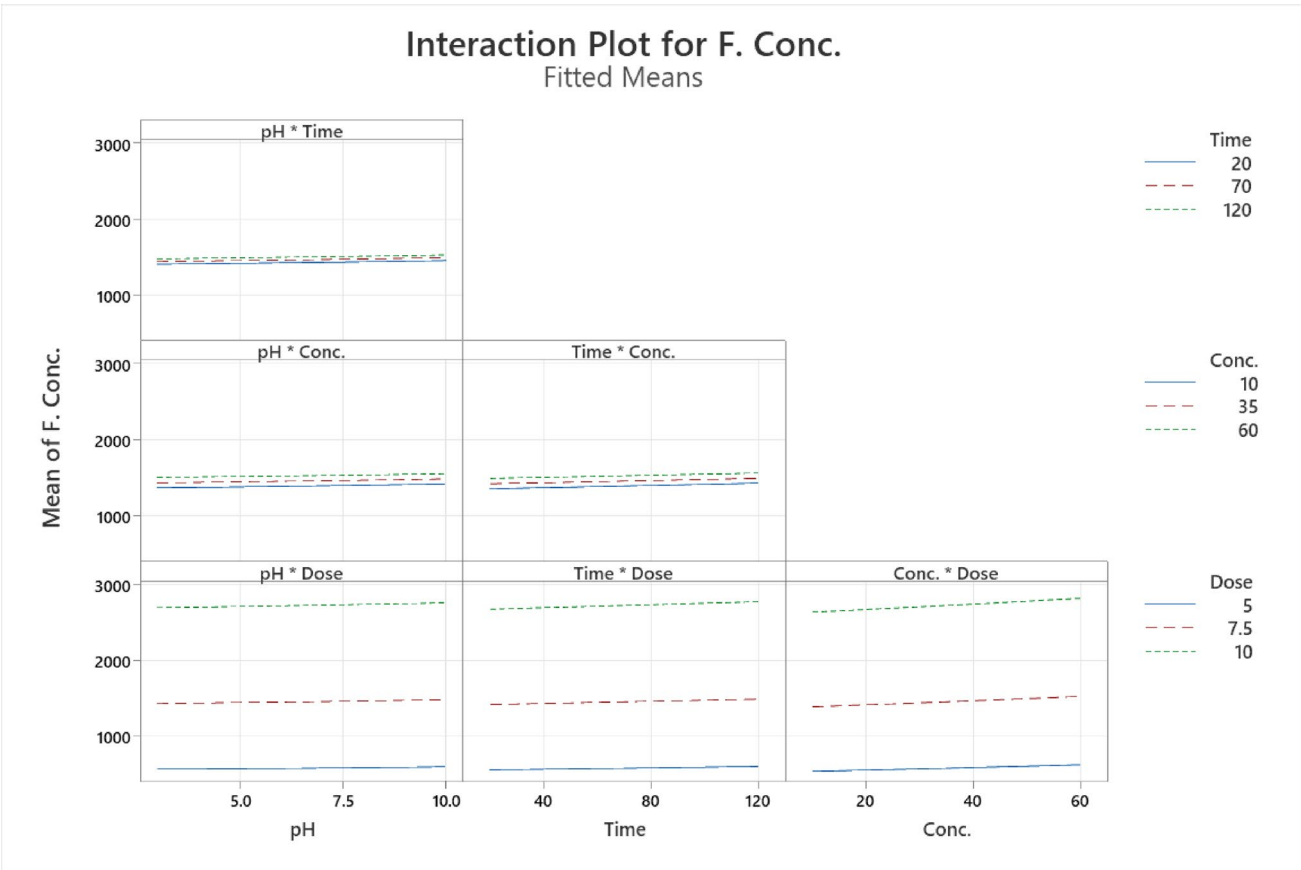

Fig. 1 Interaction effects plot for final Cr(VI) concentration (F. Conc.) depicting the Two-Way interactions among pH, contact time, initial concentration, and magnetite dose

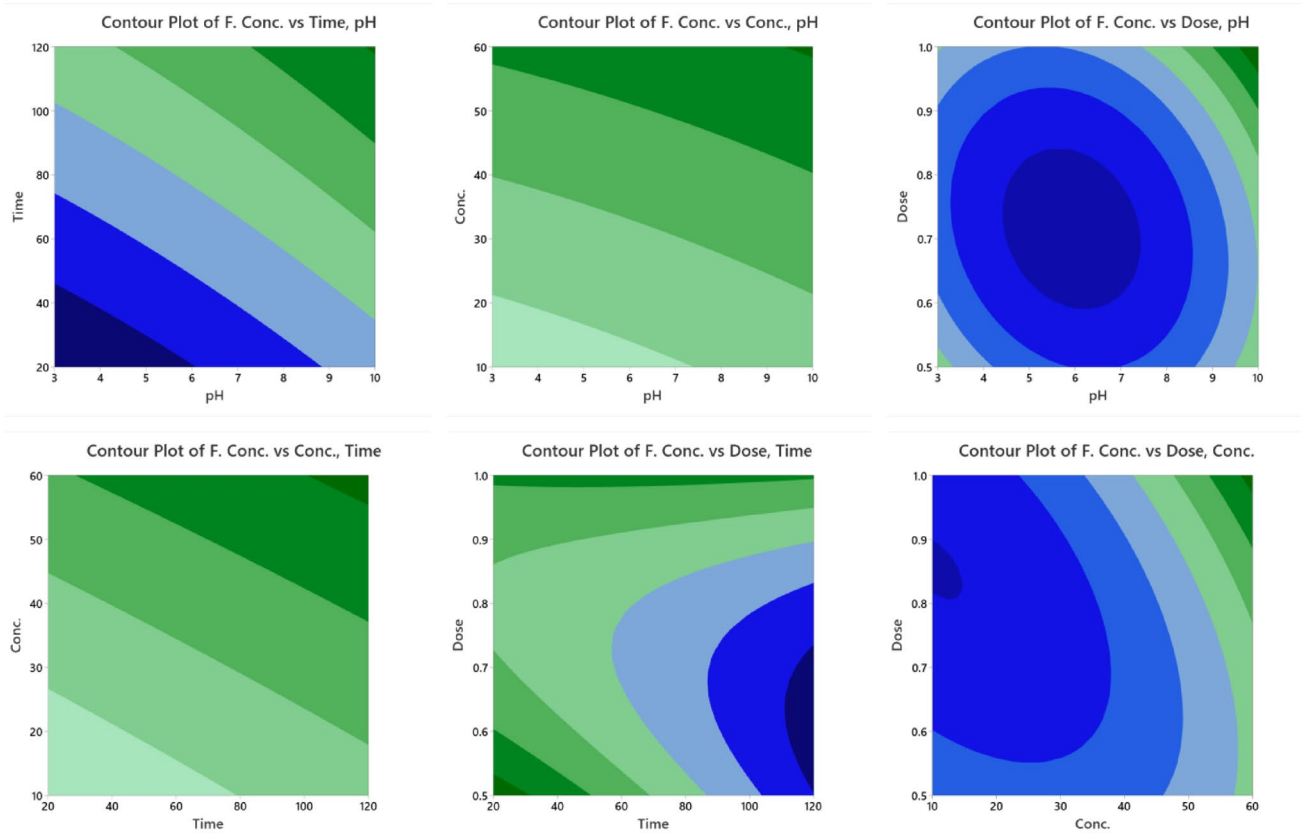

Fig. 2 Contour plots illustrating the interactive effects of key process variables: pH, contact time, initial Cr(VI) concentration, and magnetite dose on final Cr(VI) concentration (F. Conc.)

at particular combinations. For instance, low concentration and high dose produce the lowest final Cr(VI) levels, while high concentration requires proportionally higher dose to achieve comparable results.

Heatmaps and grouped bar plots presented in Fig. 4 show comparative views across multiple combinations. These visualizations highlight the critical roles of pH and dose, both are strongly significant, especially under varying initial concentrations and durations. The heatmap (a) illustrates the relationship between final Cr(VI) concentration (F. Conc.), contact time, and pH levels. Each column represents a specific F. Conc. value, while rows correspond to three distinct contact times: 20, 70, and 120 min. The color gradient, from deep blue to red, indicates the average pH associated with each time and F. Conc. combination, with blue representing low pH (acidic conditions) and red representing high pH (alkaline conditions). The plot reveals that low final Cr(VI) concentrations are predominantly associated with lower pH values, particularly at extended contact times (120 min), where more dark blue tiles are observed. This visual trend confirms that acidic pH and longer contact time favor Cr(VI) removal, aligning with the findings that low pH enhances adsorption efficiency through favorable electrostatic interactions between Cr(VI) anions and the positively charged Fe_3O_4 surface [50].

This heatmap (b) illustrates the interaction between Fe_3O_4 dose and final Cr(VI) concentration (F. Conc.) across different pH levels, as indicated by the color gradient. Each column represents a specific F. Conc. value, while rows correspond to three levels of Fe_3O_4 dose: 0.50, 0.75, and 1.00 g/L. The color shading, ranging from deep blue (low pH) to red (high pH) denotes the mean pH associated with each dose–concentration combination. The plot clearly shows that the lowest F. Conc. values are associated with moderate to high doses (especially at 0.75 and 1.00 g/L) under acidic conditions (blue shades), reflecting optimal Cr(VI) removal. Conversely, high F. Conc. values are often linked with higher pH levels (red shades), particularly at lower doses (0.50 g/L), where removal performance diminishes. This pattern reinforces the significance of both dose and pH in achieving efficient Cr(VI) adsorption, emphasizing that an adequate magnetite dose under acidic conditions is essential for minimizing residual Cr(VI) in solution.

Optimization analysis using the desirability function approach identified the optimal operating conditions required to achieve a target final Cr(VI) concentration (F.

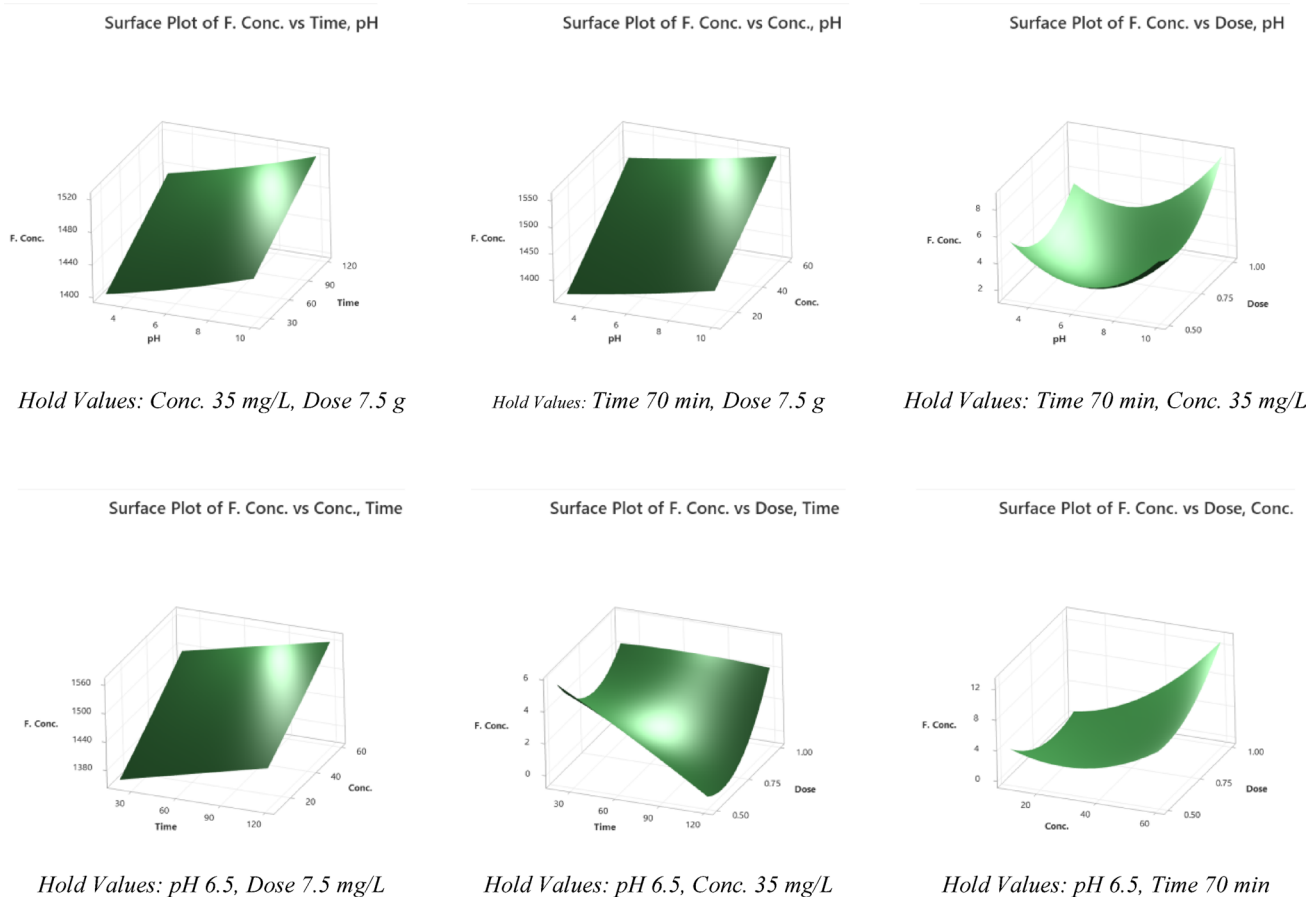

Fig. 3 Three-dimensional surface plots showing the effects and interactions of pH, contact time, initial Cr(VI) concentration, and magnetite dose on the final Cr(VI) concentration (F. Conc.)

Conc.) of 0.001 mg/L, effectively purposing near-complete removal. The optimal conditions were determined to be: pH = 3.0, contact time = 21.64 min, initial Cr(VI) concentration = 12.84 mg/L, and Fe_3O_4 dose = 1 g/L. Under these conditions, experimental validation yielded a final Cr(VI) concentration of 0.001116 mg/L, closely matching the predicted value. Therefore, Cr(VI) removal using Fe_3O_4 is governed by complex but predictable interactions among pH, contact time, concentration, and dose. Fe_3O_4 dose emerges as the dominant variable, particularly in its interactions with contact time and initial concentration. The sustained performance of Fe_3O_4 can be attributed to its high surface area, abundance of Fe(II)/Fe(III) redox-active sites, and strong adsorption capacity. These properties not only facilitate efficient Cr(VI) binding through electrostatic attraction and surface complexation but also enable partial reduction of Cr(VI) to the less toxic Cr(III), enhancing the long-term adsorption capacity [11, 12]. On the other hand, while pH has a limited standalone effect, it contributes more meaningfully through quadratic and interactive terms.

Despite the significant performance of Fe_3O_4 in removing Cr(VI) from aqueous solutions, this research emphasizes the valorization of waste materials, eggshells, to develop sustainable, low-cost alternatives that achieve comparable or even superior removal efficiency. By integrating eggshells with Fe_3O_4 (ESM) to form a hybrid eco-friendly biocomposite, the study not only enhances adsorption capacity through synergistic interactions but also reduces reliance on pure chemical reagents, aligning with environmental sustainability and circular economy principles.

Adsorption performance and kinetic behavior of ESM biocomposites for Cr(VI) removal

Figure 5 illustrates the Cr(VI) removal efficiency over time using ESM biocomposites prepared at different eggshell-to-magnetite weight ratios: 2:1, 1:1, 4:1, and 5:1 (wt%). These ratios represent the relative mass of eggshell (a low-cost, calcium-rich biosorbent) to Fe_3O_4 nanoparticles (a highly reactive component offering proved adsorption performance). Contact times of 15, 40, 60, and 120 min were evaluated to observe both rapid and equilibrium-phase adsorption behavior.

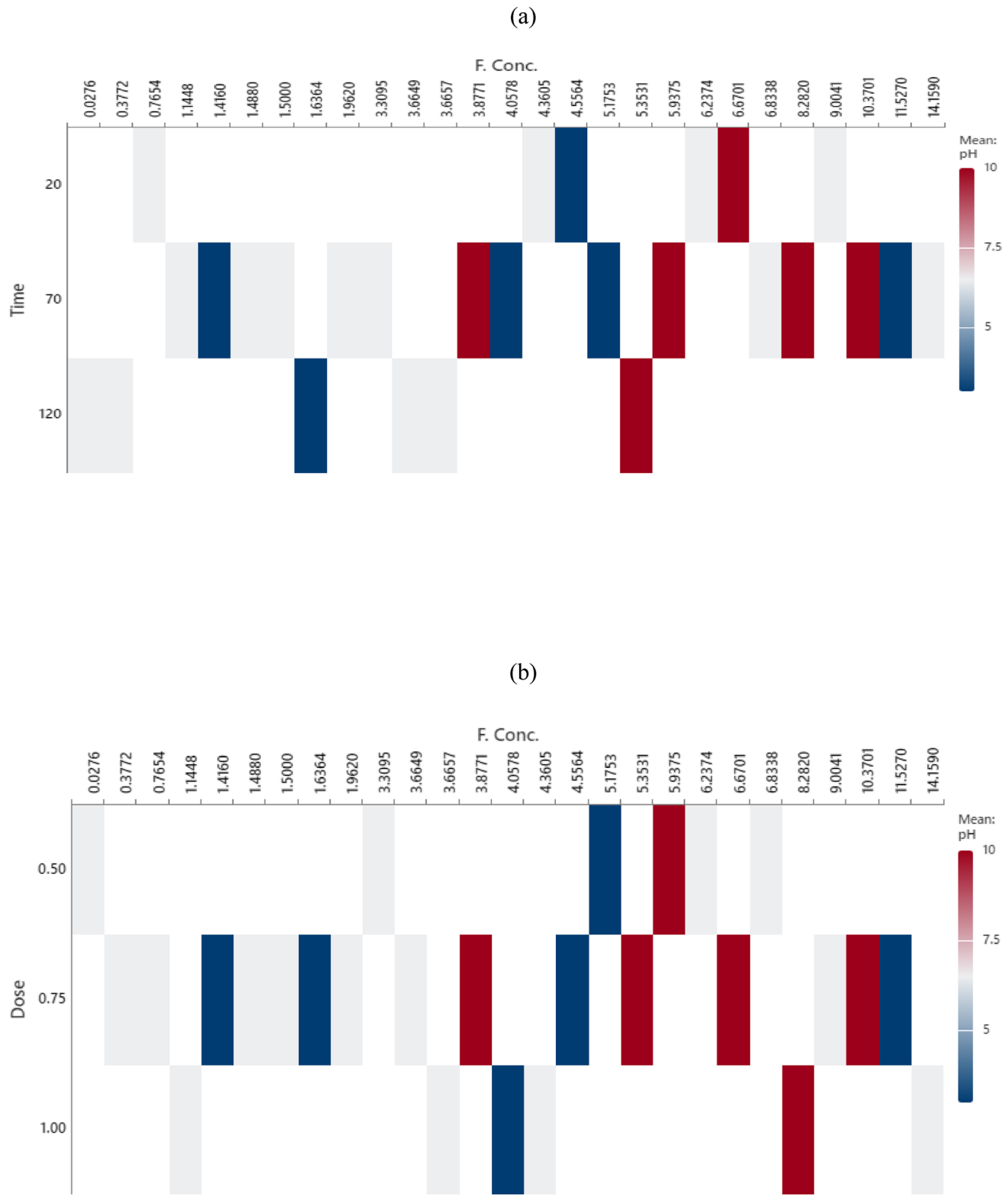

Fig. 4 Heatmaps and categorical interaction plots on the final Cr(VI) concentration (F. Conc.): **a** Heatmap of final Cr(VI) concentration as a function of contact time and pH, **b** Heatmap of final Cr(VI) concentration as a function of magnetite dose and pH

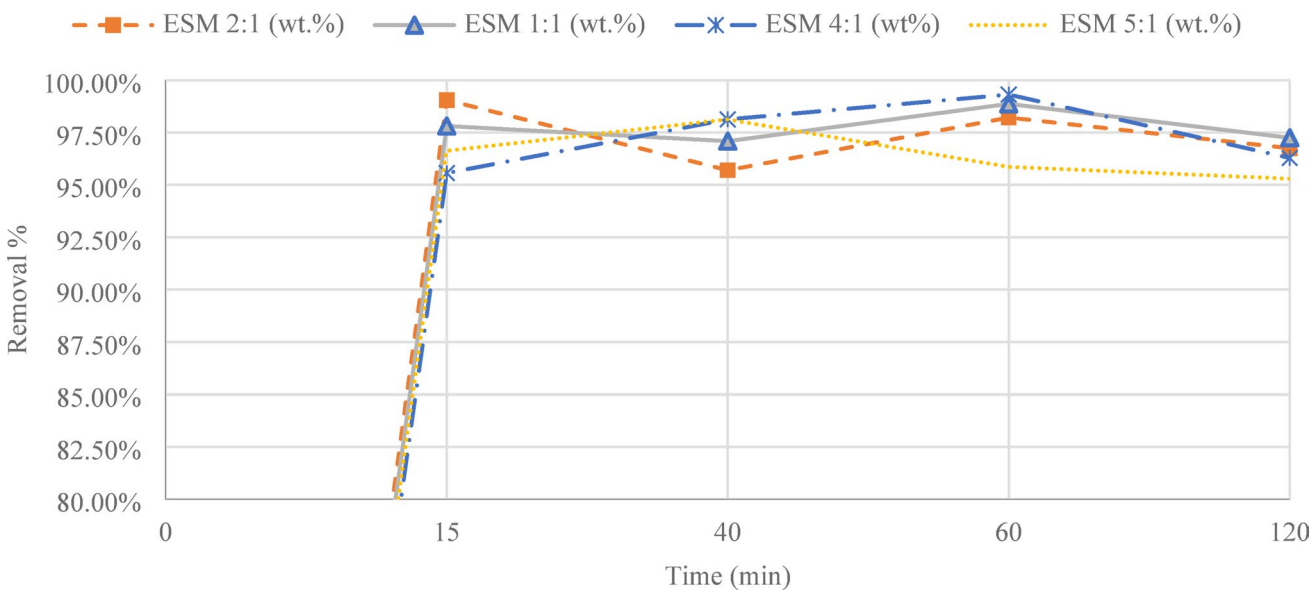

Fig. 5 Effect of contact time on Cr(VI) removal efficiency using ESM biocomposites at various ratios (1:1, 2:1, 4:1, 5:1 wt%); the ESM 2:1 wt% exhibited the highest and fastest removal efficiency (99.05% at 15 min, pH 3.0)

Table 6 Comparative performance of ESM and related literature composites for Cr(VI) removal

Adsorbent	Optimal pH	Contact time (min)	Max. Cr(VI) Removal (%)	Operational parameters and observations	References
ESM 2:1	3.0	15	99.05	Rapid, high efficiency, low magnetite content	This study
ESM 4:1		60	~ 99	Higher at longer time, slower initial uptake	
ESM 5:1		15–120	< 95	Aggregation, reduced efficiency	
Fe_3O_4 only		21.64	12.84	Super performance at optimized conditions	
Eggshell only		15/120	90 / 70	Declines over time, desorption observed	
Eggshell- Fe_3O_4	2.0	30	95	Industrial effluent, exothermic adsorption	Ajaba et al. [14]
Eggshell-Fe_3O_4 nanoadsorbent	5.5	66.2	92.9	94.5 mg/L Cr(VI)	R.T. & Sundararaman [17]
Eggshell/polypyrrole composite	5.0	30	> 80	Polyethylene glycol, water/ethanol solvent	Ehsanpour et al. [18]
Ethiopian low-cost Odaracha adsorbent (Local clay)	3.0	180	94.68	Lower performance due to much higher initial Cr(VI) concentration (110 mg/L)	Birhanu et al. [21]
Fe_3O_4 (electrochemically enhanced)	2.0	–	93.7	Electrochemical reduction, 514.7 mg/g capacity	Yang et al. [22]
Fe_3O_4/pyrrhotite composite	–	–	100	Electrokinetic, stable immobilization	Zhou et al. [23]

Whatever the mixing ratio, all ESM experiments achieved high adsorption performance with Cr(VI) exceeds 95% after 15 min. Among the tested formulations, the ESM 2:1 composite consistently demonstrated the most effective and balanced performance across all contact times. A key observation is the efficient early performance of the ESM 2:1 biocomposite, which achieved the highest removal efficiency (nearly 90.05%) within 15 min, which was largely driven by external surface binding and initial mass transfer as evidenced by characterization analysis. Although the ESM 4:1 ratio showed slightly higher performance at 60 min (~99%), its initial removal efficiency at 15 min was lower than that of the 2:1 ratio. This suggests that while 4:1 may be advantageous for longer contact durations, the 2:1 ratio provides a superior balance between speed and efficiency; making it highly suitable for real scenarios where rapid treatment is essential.

On the other hand, the consistent but slightly lower performance of the 1:1 ratio and the relatively poor performance of the 5:1 ratio (with excessive eggshell content) further reinforces the importance of optimizing the eggshell-to-magnetite balance. While eggshells offer a high initial surface area and porosity due to their calcium carbonate matrix, they lack the functional groups and redox-active sites necessary

for sustained Cr(VI) uptake [43]. Additionally, desorption and equilibrium back-diffusion may occur more readily in eggshell-only systems. This is primarily due to the nature of the adsorption mechanism and the lack of strong binding or redox-active sites on raw eggshells [51]. In the case of 5:1, eggshell aggregation may limit active surface availability in the biocomposite, negatively affecting removal efficiency. A high eggshell content can lead to poor dispersion of the functional modifier throughout the composite. This results in fewer redox-active or functional groups being exposed on the surface, further limiting adsorption and reduction of Cr(VI) [17, 18, 52].

As depicted in Fig. 5, the eggshell material shows a distinct behavior compared to the ESM composites containing Fe_3O_4. At the initial time point (15 min), the eggshell alone achieves a moderate removal efficiency of around 90%, indicating that eggshell has some inherent capacity to adsorb Cr(VI) due to the presence of functional groups such as carbonate and hydroxyl groups in its calcium carbonate structure [48]. However, beyond the initial phase, the removal efficiency of the eggshell decreases substantially with time, dropping to about 70% after 120 min. This suggests that while the eggshell surface offers initial adsorption sites for Cr(VI), it lacks the stability and reactivity necessary for sustained adsorption or possible reduction of Cr(VI) to Cr(III) [53]. This also confirms the former interpretation that, as the system approaches equilibrium, some desorption or back-diffusion of Cr(VI) ions from the eggshell surface likely occurs, resulting in the observed decline in removal efficiency. The observed decline in performance over time underscores the essential role of Fe_3O_4 in stabilizing the adsorbed Cr(VI) and enhancing long-term treatment efficacy in composite systems [54].

A comparison between the performance of pure magnetite nanoparticles (Sect. RSM experimental results for Cr(VI) removal using Fe_3O_4 Only) and the ESM 2:1 biocomposite highlights the practical benefits of composite design for Cr(VI) removal. Under similar experimental conditions, Fe_3O_4 alone achieved a slightly higher removal efficiency of 99.99% at pH 3.0 and a contact time of 21.64 min. In contrast, the ESM 2:1 composite reached a comparable 99.05% removal at the same pH but within a notably shorter duration of just 15 min with lower Fe_3O_4 content. The early phase of Cr(VI) adsorption using ESM 2:1 is driven predominantly by the enhanced concentration gradient between the solution and sorbent surface. This result underscores the effectiveness of incorporating eggshell as a support matrix, providing a rapid, efficient, and cost-effective alternative to pure magnetite.

Conversely, the ESM 2:1 composite presents a well-optimized configuration that combines the high reactivity of Fe_3O_4 with the structural support and dispersive properties of eggshell. The results demonstrate that the eggshell matrix enhances the distribution of magnetite particles, reduces their tendency to agglomerate, and contributes additional porosity that improves access to adsorption sites [55]. The elevated initial Cr(VI) uptake observed with the ESM 2:1 composite is likely due to the increased availability of these active sites, along with improved mass transfer facilitated by the eggshell structure.

When compared to findings in the literature, ESM 2:1 revealed interesting performance. For example, Ajaba et al. [14] reported a maximum of 95% Cr(VI) removal using an eggshell-magnetite composite at pH 2.0 with a contact time of 30 min, double the duration required by the ESM 2:1 to achieve even higher removal. While the reported is removal efficiency is high, it required a more acidic environment and double the contact time compared to the ESM 2:1 composite in this study, which achieved 99.05% removal at pH 3.0 in just 15 min. The need for a lower pH in Ajaba et al.'s work may increase chemical costs and limit practical application, especially for large-scale or environmentally sensitive operations. Similarly, Sundararaman et al. [17] synthesized an eggshell–Fe_3O_4 nanocomposite and achieved 92.9% Cr(VI) removal at pH 5.5, but only after 66.2 min of treatment. This result demonstrates that higher pH operation is possible, but at the expense of much slower kinetics and lower overall removal. The longer contact time required may be due to less efficient dispersion of Fe_3O_4 or lower surface reactivity, as their material characterization showed less agglomeration but not the same rapid uptake as ESM 2:1 composite. Another study by Ehsanpour et al. [18] investigated an eggshell/polypyrrole composite and reported over 80% removal at pH 5.0 in 30 min. While this system benefits from the conductive and redox-active properties of polypyrrole, the removal efficiency and kinetics are still lower than those of ESM 2:1 composite. This suggests that while polymer modification can enhance performance under milder conditions, the synergy between eggshell and magnetite in ESM composite offers a more effective balance of speed and capacity.

Thus, the ESM 2:1 composite achieves rapid and near-complete Cr(VI) adsorption under moderately acidic conditions (pH 3.0), without the need for extended contact durations. This not only enhances process feasibility but also reduces chemical usage and operational costs. While pure Fe_3O_4 in this study achieved > 99% removal at optimized conditions (pH 3.0, contact time ~ 21.6 min), it required fine-tuned operational parameters, which may be difficult to maintain in real-world applications. Moreover, Fe_3O_4 synthesis and use at higher doses can be cost-prohibitive. The ESM 2:1 composite, on the other hand, offers a more cost-effective and robust alternative, leveraging the structural and dispersive advantages of waste eggshells to

enhance Fe_3O_4 performance while reducing its required quantity.

Adsorption isotherm and kinetics

The adsorption isotherm describes how Cr(VI) ions interact with an adsorbent surface at equilibrium, revealing important information about the adsorption mechanism and the nature of the adsorbent. Two widely used models to analyze adsorption data are the Freundlich and isotherms (Eqs. 4 and 5), respectively. Each provides a different perspective on how Cr(VI) is captured from solution.

$$\frac{X}{M} = kC^{\frac{1}{n}} \tag{4}$$

$$\frac{X}{M} = \frac{abC}{1+bC} \tag{5}$$

where: X = adsorbate adsorbed by adsorbent ($X = C_0 - C$), mg adsorbate/L, M = carbon dose (g/L) as activated carbon, X/M = mass of adsorbate adsorbed per unit mass of adsorbent (mg adsorbate/g) activated carbon, C_0 = initial concentration of adsorbate, mg/L, C = final or equilibrium concentration of the adsorbate, mg/L, k = empirical constant or Freundlich capacity factor (mg adsorbate/g) activated carbon. The ln(k) is the Y-intercept when ln(X/M) is plotted with respect to ln(C), 1/n = slope constant or Freundlich intensity parameter, dimensionless. It is the slope on the plot of ln(X/M) versus ln(C), a = maximum mass of adsorbate adsorbed per unit mass of adsorbent (mg adsorbate/g) activated carbon, and b = empirical constant (L/mg) activate carbon [56].

The Freundlich isotherm model assumes that adsorption occurs on a heterogeneous surface, meaning that different sites have different energies of adsorption. This model allows for multilayer adsorption, where Cr(VI) ions can form layers on top of each other. As shown in Fig. 6a, the Freundlich model is characterized by two parameters: the Freundlich constant (k = 7.43 mg/g, which indicates adsorption capacity) and the intensity parameter ($1/n = 0.9739$), which reflects how favorable the adsorption is. A value of 1/n less than 1 suggests that the adsorption process is favorable and that the surface is energetically heterogeneous, $R^2 = 0.9732$.

In contrast, the Langmuir isotherm model assumes that adsorption occurs as a monolayer on a homogeneous surface with a finite number of identical sites. According to the model depicted in Fig. 6b, once a site is occupied by a Cr(VI) ion, no further adsorption can take place at that site, and there is no interaction between adsorbed ions. A key parameter in the Langmuir model is the maximum adsorption capacity (q_m=90.09 mg/g), which represents the maximum amount of Cr(VI) that can be adsorbed per unit mass of the adsorbent ($R^2 = 0.8832$). The Langmuir model is especially useful for determining whether the adsorption process is favorable, based on the calculated separation factor (R_L), which is estimated at 0.444, Eq. (6). When R_L falls between 0 and 1, the adsorption is considered favorable, which is often the case for Cr(VI) adsorption onto various adsorbents.

$$R_L = \frac{1}{1+bC_0} \tag{6}$$

When applying these models to the adsorption of Cr(VI) on the ESM 2:1 (wt%), experimental data fitted both isotherms well, indicating that both monolayer and multilayer adsorption occurred, depending on the experimental conditions. This finding aligns with previous literature [57], which reported high Cr(VI) removal efficiency using PAN-CNT/TiO_2 (polyacrylonitrile nanofibers embedded with carbon nanotubes and titanium dioxide nanoparticles) composite nanofibers. The enhanced performance was attributed to the nanofibers' large surface area, diverse functional groups, and porous architecture, all of which offer numerous and accessible binding sites for Cr(VI) adsorption. In the current study, a strong fit to the Freundlich model indicates a heterogeneous surface and the dominance of multilayer adsorption, while a good fit to the Langmuir model suggests that monolayer adsorption is possible. In practical terms, this means that Cr(VI) removal from aqueous solution using ESM is generally efficient, with the process being

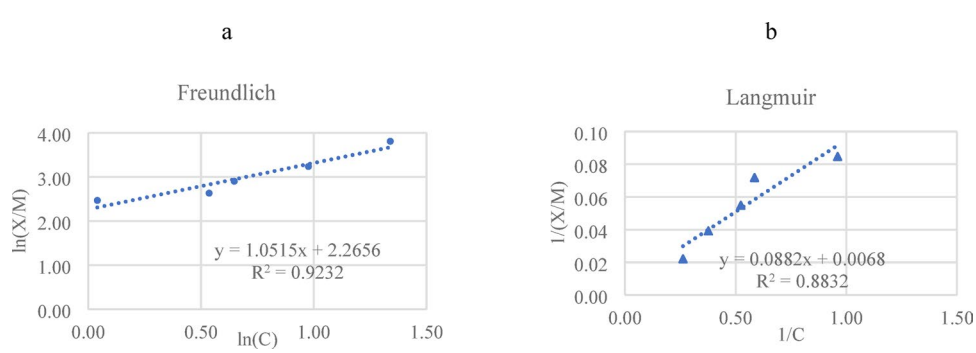

Fig. 6 Freundlich and Langmuir models of Cr(VI) ions on ESM 2:1 (wt%) adsorbent

both favorable and capable of achieving significant removal capacities.

Kinetic studies are essential for understanding the rate at which adsorption occurs and for determining the time required to reach equilibrium. In adsorption processes, the two most commonly applied kinetic models are the pseudo-first-order and pseudo-second-order models, each representing different assumptions about the mechanism of adsorption.

The pseudo-first-order kinetic model (Eq. 7), originally proposed by Lagergren, assumes that the rate of occupation of adsorption sites is directly proportional to the number of unoccupied sites in a linearized form. The pseudo-second-order kinetic model (Eq. 8) assumes that the rate-limiting step may involve chemisorption, where valency forces through sharing or exchange of electrons are involved between adsorbent and adsorbate. Where Q_e is the amount of adsorbate adsorbed at equilibrium (mg/g), Q_t is the amount adsorbed at time t (mg/g), and k_1 and K_2 are the pseudo-first-order rate constant (1/min) and the pseudo-second-order rate constant (g/mg·min), respectively.

$$\mathrm{Ln}\,(Q_e - Q_t) = -k_1 t + \ln Q_e \tag{7}$$

$$t/Q_t = \frac{t}{Q_e} + \frac{1}{k_2 Q_e^2} \tag{8}$$

In the present study, the adsorption kinetics of Cr(VI) onto ESM 2:1 (wt%) was investigated (Table 7). To determine the most appropriate kinetic model, the experimental data were fitted to both pseudo-first-order and pseudo-second-order models. The results showed that the correlation coefficient (R^2) for the pseudo-first-order model was 0.4099 (Fig. 7a), whereas the pseudo-second-order model yielded a much higher R^2 value of 0.9998 (Fig. 7b). This indicates that the adsorption process is solely described by the pseudo-second-order model, reflecting that chemisorption is the dominant mechanism. Furthermore, the adsorption capacity calculated from the pseudo-second-order model (Q_e=127.19 mg/g). This kinetic behavior is consistent with many advanced adsorbents for Cr(VI) removal [58–60].

Ajab et al. [14] developed a magnetized eggshell biosorbent by integrating Fe_3O_4 nanoparticles with Gallus gallus domestics eggshells. Their study demonstrated significant Cr(VI) removal efficiency, attributing the performance to the synergistic effect of the eggshell matrix and Fe_3O_4 nanoparticles. The adsorption process followed pseudo-second-order kinetics and conformed to the Langmuir isotherm model, indicating monolayer adsorption on a homogeneous surface. Accordingly, the synergy between these components significantly improves both the kinetics and capacity of Cr(VI) adsorption.

Regeneration and reusability of the ESM biocomposite

A critical factor in evaluating the practical applicability of any adsorbent is its reusability after regeneration, which directly influences operational cost, sustainability, and long-term performance in real-world wastewater treatment scenarios. To assess this, the ESM (2:1 wt%) composite (identified as the most effective formulation from earlier batch studies) was subjected to three sequential Cr(VI) adsorption cycles, with measurements taken at four contact times (15, 40, 60, and 120 min) per cycle, employing operating conditions of pH 3.0, initial concentration 12.84 mg/L, and ESM dose at 1.0 gm.

As presented in Table 8, the original experiment using fresh ESM 2:1 composite achieved an outstanding removal efficiency of 99.05% at 15 min, underscoring its high initial capacity for Cr(VI) removal (Sect. Adsorption performance and kinetic behavior of ESM biocomposites for Cr(VI) removal). After the first regeneration cycle, the ESM 2:1 composite maintained a robust performance, with removal efficiencies ranging from 96.73% at 15 min to 93.69% at 120 min. This slight decline is typical in adsorption recycling experiments and is mainly due to partial saturation of adsorption sites or minor structural changes that occur during the desorption phase. Interestingly, the data also suggest that some performance improvement may arise from the activation of previously blocked or inaccessible adsorption sites during the initial adsorption. Such behavior has been observed in other bio-inorganic hybrid materials, where

Table 7 Experimental data for the adsorption of Cr(VI) from aqueous solution (initial concentration = 12.84 mg/L) onto ESM (2:1 wt%) at varying adsorbent doses, including calculated parameters for Freundlich and Langmuir isotherms

ESM (2:1 wt%) dose (g)	C_o (mg/L)	C_e (mg/L)	$X = C_o - C_e$ (mg/L)	X/M (mg/g)	Freundlich isotherm		Langmuir isotherm		
					ln(C)	ln(X/M)	C/(X/M) (mg/L)	1/C (L/mg)	1/(X/M) (mg/mg)
0.2	12.84	3.817	9.023	45.12	1.34	3.81	0.08	0.26	0.02
0.4	12.84	2.655	10.185	25.46	0.98	3.24	0.10	0.38	0.04
0.6	12.84	1.9104	10.9296	18.22	0.65	2.90	0.10	0.52	0.05
0.8	12.84	1.7091	11.1309	13.91	0.54	2.63	0.12	0.59	0.07
1	12.84	1.041	11.799	11.80	0.04	2.47	0.09	0.96	0.08

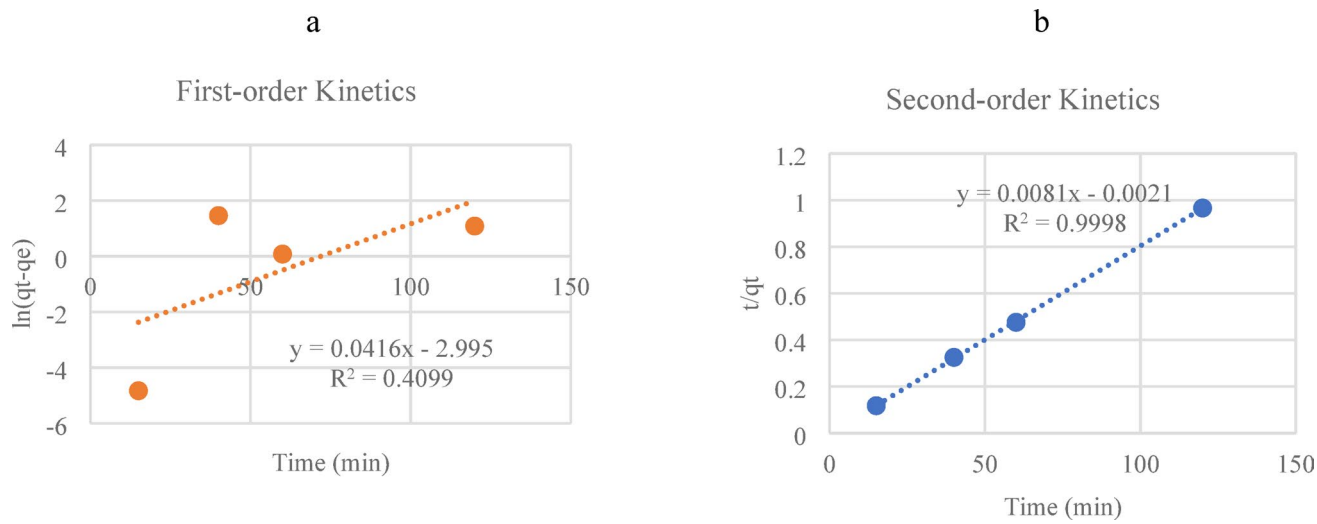

Fig. 7 Pseudo 1st and 2nd order kinetics of Cr((VI) on ESM 2:1 (wt%)

Table 8 Residual Cr(VI) concentration, removal efficiency (R%), and remaining adsorbent mass of the ESM (2:1 wt%) composite across three regeneration cycles at various contact times

Original Experiment	Time (min)	Concentration (mg/L)				R%				Wt. at the cycle end (gm)
		0.1218 After 15 min				99.05% After 15 min				
Cycles		15	40	60	120	15	40	60	120	
After regeneration	Cycle 1	0.420	0.621	0.826	0.810	96.73%	95.17%	93.57%	93.69%	0.96
	Cycle 2	0.831	0.921	1.386	0.962	93.53%	92.83%	89.21%	92.51%	0.955
	Cycle 3	1.318	1.371	1.790	2.103	89.74%	89.33%	86.06%	83.62%	0.84

repeated cycling can temporarily enhance active site accessibility and surface reactivity [61], making the performance deterioration limited.

The second regeneration cycle demonstrated a more noticeable decrease in performance, with removal efficiencies spanning 93.53% at 15 min down to 89.21% at 60 min. This trend suggests that repeated use progressively diminishes the number of active sites available for Cr(VI) adsorption, likely due to pore blockage or structural fatigue. This stability may be partly explained by the BET and BJH data presented in Sect. Surface textural and structural changes. After adsorption, the surface area of the composite increased from 31.34 to 84.38 m^2/g, and pore volume more than doubled, reflecting progressive activation and restructuring of the adsorbent's internal architecture. These enhancements likely remain intact through the first two regeneration cycles, contributing to sustained performance.

By the third regeneration cycle, removal efficiencies decreased further, reaching 89.74% at 15 min and dropping to 83.62% at 120 min. The residual Cr(VI) concentrations correspondingly increased, indicating that some active sites had become irreversibly occupied or that structural integrity of the composite had deteriorated. While this drop is modest, it does indicate functional fatigue, possibly due to accumulation of unreleased Cr(VI) ions on deeper or less accessible pores, minor structural breakdown or compaction within the matrix, and/or chemical alteration of reactive sites. Thus, the ESM composite demonstrates regeneration and stability trends similar to leading Cr(VI) adsorbents, with high efficiency sustained through at least two cycles due to internal activation and restructuring solutions [32, 62, 63]. This positions ESM as a competitive, reusable material for Cr(VI) removal from aqueous.

Besides, the original experiment began with a mass of 1.0 gram, which decreased slightly to 0.96 g after the first regeneration cycle and then to 0.955 and 0.84 g after the second and third cycles, respectively. This gradual mass loss, particularly notable in the third cycle, suggests a degree of material erosion or loss of active sites over repeated adsorption and regeneration processes resulted from harsh acidic condition (pH 3.0). Such a trend is commonly observed in practical biosorbent reuse scenarios, reflecting mechanical or chemical stress on the adsorbent [64]. However, the minimal overall reduction in the composite's mass across cycles also indicates that the ESM 2:1 composite retains substantial structural integrity and exhibits limited degradation. This durability reinforces its potential for multiple reuse cycles, offering a practical and sustainable approach for Cr(VI) removal in real-world water treatment applications. High regeneration performance means that the operational

lifespan of the material can be extended, reducing replacement frequency and overall treatment costs.

Economic analysis of ESM-based Cr(VI) treatment

A robust cost analysis is essential to assess the economic viability of using ESM composites for Cr(VI) removal from aqueous solutions. The detailed cost structure is prepared with estimations tailored to the Egyptian context for treating one cubic meter of contaminated water. Eggshells are an abundant food industry byproduct in Egypt, typically available at a negligible or zero cost, with expenses limited to collection, cleaning, and transportation. Magnetite can be sourced locally or synthesized, with costs depending on the method and scale. The preparation of the ESM bioadsorbent involves washing, drying, and grinding eggshells, followed by synthesis or coating with magnetite. These steps require energy, labor, and basic equipment [65].

Operation and Maintenance (O&M) costs include labor for supervision, energy for mixing/agitation, water for washing, periodic adsorbent replacement or regeneration, and safe disposal of the spent material. The total cost (TC) can be estimated by summing the cost of raw materials (C_{raw}), manufacturing ($C_{manufacturing}$), and operation & maintenance ($C_{O\&M}$), according to Eq. (9). The analysis is based on treating a flow of 1.0 m^3/day to remove Cr(VI) from an initial concentration of 12.84 mg/L to a target of 99.05% removal, using an ESM (2:1 wt%) composite.

$$TC = C_{raw} + C_{manufacturing} + C_{O\&M} \tag{9}$$

As detailed in Table 9, the raw material and manufacturing cost ($C_{raw} + C_{manufacturing}$) for synthesizing 1 gram of Fe_3O_4 nanoparticles was calculated at \$0.30/g, based on lab-scale procedures involving iron nitrate and ethylene glycol (as detailed in Sect. Synthesis of Fe_3O_4 nanoparticles via ethylene glycol-assisted method). In contrast, the eggshell powder, derived from food-industry waste through simple cleaning and drying steps, had a significantly lower cost of \$0.015/g. The ESM composite was prepared at a 2:1 weight ratio, resulting in an average preparation cost of \$0.109/g for the composite.

To achieve optimal Cr(VI) removal efficiency (99.05%) at an initial Cr(VI) concentration of 12.84 mg/L, the required dosage of ESM 2:1 was estimated using both estimated Freundlich and Langmuir isotherms (Sect. Adsorption isotherm and kinetics), yielding an average requirement of 0.01265 kg/day of composite. This corresponds to approximately 0.0042 kg/day of magnetite and 0.0084 kg/day of eggshell. The calculated yearly fixed capital expenditure (FCE) associated with material synthesis and usage was estimated at \$1.386/m^3/day.

Operational and maintenance ($C_{O\&M}$) costs were also assessed, considering local labor rates in Egypt (approximately \$0.48/h) and typical daily equipment maintenance and depreciation costs of \$6.00/day. This resulted in a yearly $C_{O\&M}$ cost of approximately \$9.84/m^3/day. Summing both fixed and operational expenses, the total treatment cost (TC) using the ESM 2:1 biocomposite was estimated at \$11.23 per cubic meter per day, demonstrating the material's economic viability for real-world water treatment applications.

The above cost estimations are highly variable and can fluctuate significantly based on local factors such as labor wages, energy prices, availability of materials, and infrastructure conditions.

Table 9 Estimated cost breakdown for synthesizing ESM (2:1 wt%) composite and treating 1.0 m^3/day of Cr(VI)-contaminated water

Cost category	Description	Value	Unit
Raw material costs ($C_{raw} + C_{manufacturing}$)			
Fe_3O_4 nanoparticles	Lab-scale synthesis (10–50 nm)	0.30	\$/g
Eggshell powder	Processed from waste eggshells	0.015	\$/g
Total ESM 2:1 biocomposite cost	Biocomposite ratio: 2:1 (wt%)	0.109	\$/g
Required ESM 2:1 amount	Based on average isotherm capacity, Sect. Adsorption isotherm and kinetics	0.01265	kg/day
Required Fe_3O_4	For composite formulation	0.0042	kg/day
Required eggshell	For composite formulation	0.0084	kg/day
Estimated daily FCE	For material procurement	1.386	\$/m^3/day
$C_{O\&M}$			
Labor (handling and supervision)	Based on typical rates in Egypt (8 h/d)	0.48	\$/h
Dailu depreciation and maintenance	Equipment O&M	6.00	\$/day
Estimated daily $C_{O\&M}$	Total operational cost	9.84	\$/m^3/day
Total cost (TC)	FCE + $C_{O\&M}$	11.23	\$/m^3/day

Discussion

Interdependent effects and adsorption behavior of the investigated adsorbents during Cr(VI) removal

The comprehensive experimental investigation into the Cr(VI) adsorption process using different investigated adsorbents revealed significant results regarding both mechanistic understanding and practical applicability. The process parameters (particularly Fe_3O_4 dosage, pH, contact time, and initial Cr(VI) concentration) were proved to play

interdependent roles in shaping adsorption performance, and these effects were statistically validated through RSM-based regression modeling for Fe_3O_4 adsorbent. The developed model (R^2 = 96.09%) clearly identified Fe_3O_4 dose as the most influential factor, exhibiting a strong negative linear coefficient (− 72.1) and a high t-value (T = 4.39, p = 0.001), confirming its critical role in controlling redox and sorption capacity. This is aligned with previous reports that emphasize the dominance of Fe_3O_4 in Cr(VI) removal, although the quadratic relationship also warns against excess dosing due to nanoparticle agglomeration, which can reduce surface area and accessibility [13].

The solution pH was also highly significant, particularly the quadratic term (pH^2, $p<0.000$), and multiple interactions involving pH were detected (e.g., pH × dose, pH × concentration). The optimal removal occurred at pH≈3.0, which is consistent with established Cr(VI) removal mechanisms where the surface of Fe_3O_4 becomes protonated under acidic conditions, thereby facilitating electrostatic attraction of anionic chromate (CrO_4^{2-}) and and dichromate ($Cr_2O_7^{2-}$) species. These observations are consistent with widely accepted findings across various adsorbents, where acidic pH (typically 2–4) enhances electrostatic attraction and reduction processes during Cr(VI) uptake. This trend is robust across a wide range of adsorbent materials and is a key parameter in process optimization. Moreover, significant interaction effects (overall $p=0.003$) were observed, particularly for dose × concentration and dose × time ($p=0.001$ and $p=0.022$, respectively). These findings reinforce the importance of balancing initial contaminant levels with sufficient adsorbent mass and appropriate contact time. Optimization via the desirability function confirmed that the best removal was achieved at pH=3.0, dose=1.0 g/L, contact time≈21.6 min, and initial Cr(VI)=12.84 mg/L, achieving nearly complete removal (predicted: 0.001 mg/L; actual: 0.001116 mg/L).

In terms of comparative performance, while pure Fe_3O_4 achieved > 99.9% removal under ideal obtained conditions, the ESM 2:1 biocomposite achieved 99.05% removal in only 15 min with less magnetite content. The incorporation of waste-derived eggshells not only enhances the dispersion and structural stability of Fe_3O_4 but also introduces additional porosity and surface functional groups, as confirmed by characterization analyses. These findings are coincided with earlier reports that biogenic calcium carbonate matrices can act as effective scaffolds for nanomaterials, reducing agglomeration and improving accessibility of active sites [14, 17, 18]. BET data showed a 2.7-fold increase in surface area of the ESM 2:1 composite post-adsorption (from 31.3 to 84.4 m^2/g), indicating adsorption-induced pore opening and structural activation [32]. FTIR spectra revealed the involvement of hydroxyl, carbonate, and Fe–O functional groups in Cr(VI) binding, confirming a dual mechanism of surface complexation and redox interaction. XRD patterns before and after adsorption indicated partial amorphization without crystalline degradation, underscoring the composite's structural stability and reinforcing its potential for reusability. These findings are consistent with trends seen in other high-performing, structurally adaptive adsorbents [16, 66].

Kinetic analysis revealed a strong fit to the pseudo-second-order model (R^2 = 0.9998), indicating chemisorption and possible redox processes as the dominant mechanisms. Isotherm modeling confirmed both monolayer and multilayer adsorption behaviors through fits to the Langmuir (q_m = 90.09 mg/g) and Freundlich (R^2 = 0.9732, k=7.43 mg/g) models. These findings are indicative of high uptake capacity on heterogeneous surfaces and align with known Cr(VI) reduction mechanisms where Fe(II) centers in Fe_3O_4 contribute to the reduction of Cr(VI) to less toxic Cr(III), which is then immobilized.

Figure 8 illustrates the adsorption process where an ESM biocomposite is used to treat water contaminated with Cr(VI). Initially, the composite material is dispersed in the contaminated water. Over time and under specific conditions, the toxic Cr(VI) ions are adsorbed onto the surface of the composite. Once adsorbed, the Cr(VI) is reduced to the less toxic Cr(III) form. This reduction is primarily driven by the Fe(II) in the Fe_3O_4, with potential contributions from the eggshell membrane as well. The final result is the effective removal of dissolved Cr(VI) from the water, as the chromium remains bound to the composite in its less harmful Cr(III) state.

Scalability, cost, and sustainability

The promising laboratory-scale performance of the ESM 2:1 biocomposite was further evaluated in terms of scalability, cost-effectiveness, and environmental sustainability. In terms of scalability, the synthesis process involves readily scalable steps: simple mixing, drying, and calcination at moderate temperatures without the need for specialized or hazardous reagents. All materials needed are widely available in the local context. Eggshells, a common agro-industrial waste, could be sourced from bakeries and restaurants, while iron salts used for magnetite synthesis are affordable and accessible. The process does not require high-pressure, cryogenic, or inert gas systems, allowing for straightforward scale-up using basic laboratory or pilot plant infrastructure. Moreover, the use of ethylene glycol as a solvent in the magnetite synthesis step aligns with common practices and avoids highly toxic or environmentally persistent solvents.

From an economic standpoint, detailed cost analysis (Table 8) revealed that the total cost of treating 1 m^3/day

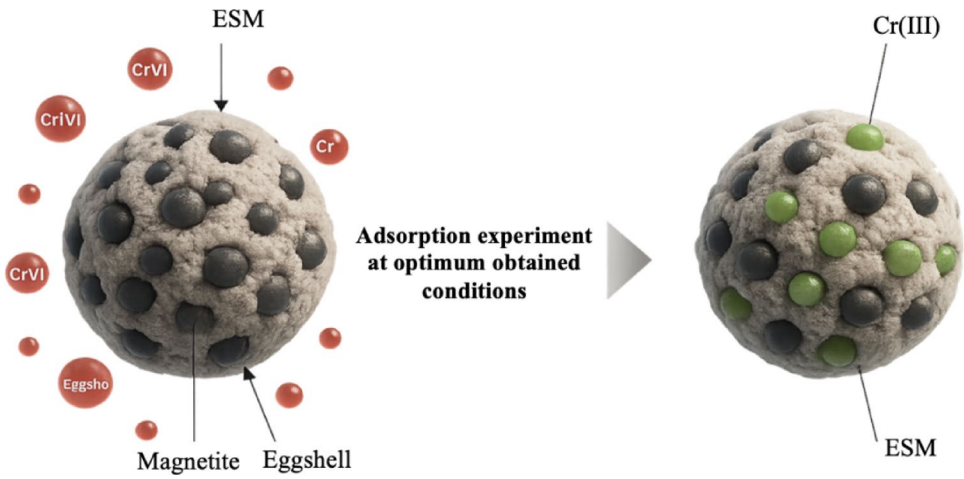

Fig. 8 Schematic illustration of Cr(VI) adsorption and reduction on the ESM (2:1 wt%) biocomposite, showing electrostatic attraction of Cr(VI) species to positively charged sites, their reduction to Cr(III) on adsorption sites, and subsequent immobilization within the ESM matrix under optimum conditions

of Cr(VI)-contaminated water using ESM 2:1 is approximately USD 11.23 per m^3/day. This includes raw material sourcing, manufacturing, and O&M costs. Raw material cost is minimized due to the use of eggshell waste and small amounts of magnetite, while labor and depreciation costs are consistent with local wage and equipment norms. The transparency of this cost breakdown and its alignment with decentralized treatment needs support the economic feasibility of real-world deployment, especially in small-scale or rural wastewater treatment scenarios.

Environmental sustainability is also a central feature of this material. Eggshells, often discarded as food waste, are repurposed into a high-value functional material, thereby contributing to circular economy principles and reducing waste sent to landfills. The composite is magnetically recoverable and showed excellent reusability over three cycles, with Cr(VI) removal efficiency remaining above 89% and only minor material degradation. Such behavior reduces the need for frequent adsorbent replacement and mitigates secondary pollution. Importantly, the synthesis process avoids the use of toxic solvents or additives, supporting the principles of green chemistry and minimizing environmental impact. Thus, the adsorption performance, reusability, and cost-efficiency using ESM 2:1 (wt%) reveals its potential for scalable and sustainable water treatment solutions.

Conclusions

This study successfully developed and characterized an eggshell–magnetite (ESM) biocomposite as an efficient, low-cost adsorbent for Cr(VI) removal from aqueous solutions. The optimized ESM (2:1 wt%) formulation achieved a maximum removal efficiency of 99.05%, driven by the synergistic action of $CaCO_3$ and Fe–O functional groups that enable ion exchange and redox-mediated reduction. The composite was synthesized through a simple, energy-efficient, and reproducible process utilizing waste eggshells and iron salts, aligning with circular economy principles. Economic evaluation revealed a total treatment cost of USD 11.23 m^{-3} day^{-1}, highlighting its competitiveness compared with commercial adsorbents. Environmentally, the approach upcycles agricultural waste, employs non-toxic reagents, and produces a wide recoverable and reusable material, underscoring its sustainability and scalability. Future studies should focus on pilot-scale implementation, regeneration cycles, and life-cycle assessment to validate the long-term feasibility of this biocomposite for decentralized wastewater treatment applications.

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Author contributions A.D.: Experimentation, Software, Original Draft Preparation; M.H.: Project Administration and Resources; A.A: Experimentation, Methodology, Supervision, and Review; A.A and A.A: Draft Review; A.F.: Conceptualization, Methodology, Supervision, and Final Review. All the authors review the manuscript.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Informed consent The current study did not involve any human or animal subjects. Therefore, it was not subject to review by an ethics committee and did not require informed consent.

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